World Happiness Report Analysis

1. Introduction

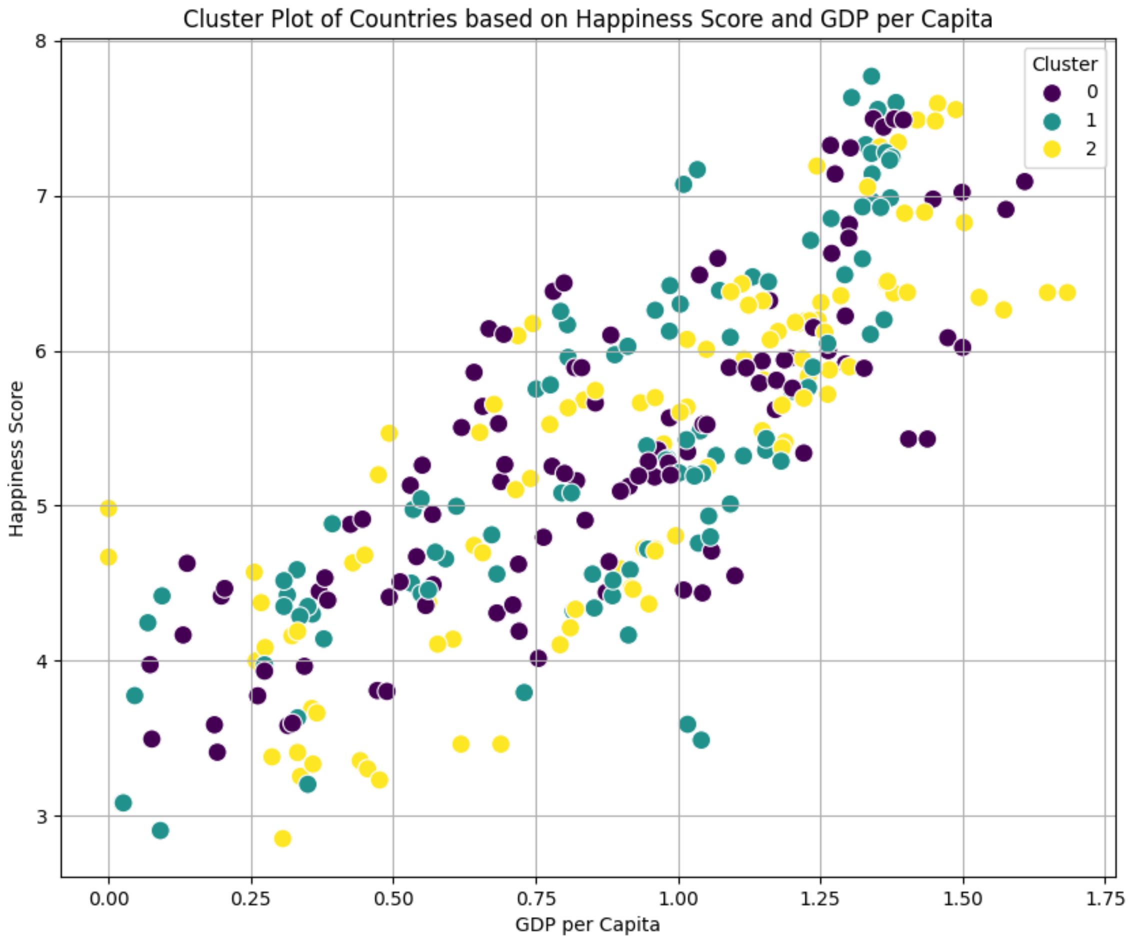
This report provides a comprehensive analysis of the World Happiness Report dataset. The analysis includes data cleaning, clustering, association rule mining, pattern analysis, and visualization to uncover insights into global happiness trends.

2. Data Cleaning

The data was cleaned by removing duplicates, handling missing values, and encoding categorical varieties. This process ensures that the dataset is ready for further analysis and modeling.

3. Clustering Analysis

K-means clustering was applied to segment countries based on their happiness scores. This helps in identifying groups of countries with similar characteristics in terms of happiness.



4. Association Rule Mining

Association rule mining was conducted to find interesting associations between different features such as GDP, social support, and life expectancy with the overall happiness score.

5. Pattern Analysis

Pattern analysis was performed to identify the most influential features contributing to high happiness scores. This includes analyzing the mean happiness score per cluster and the distribution of countries in each cluster.

6. Visualizations

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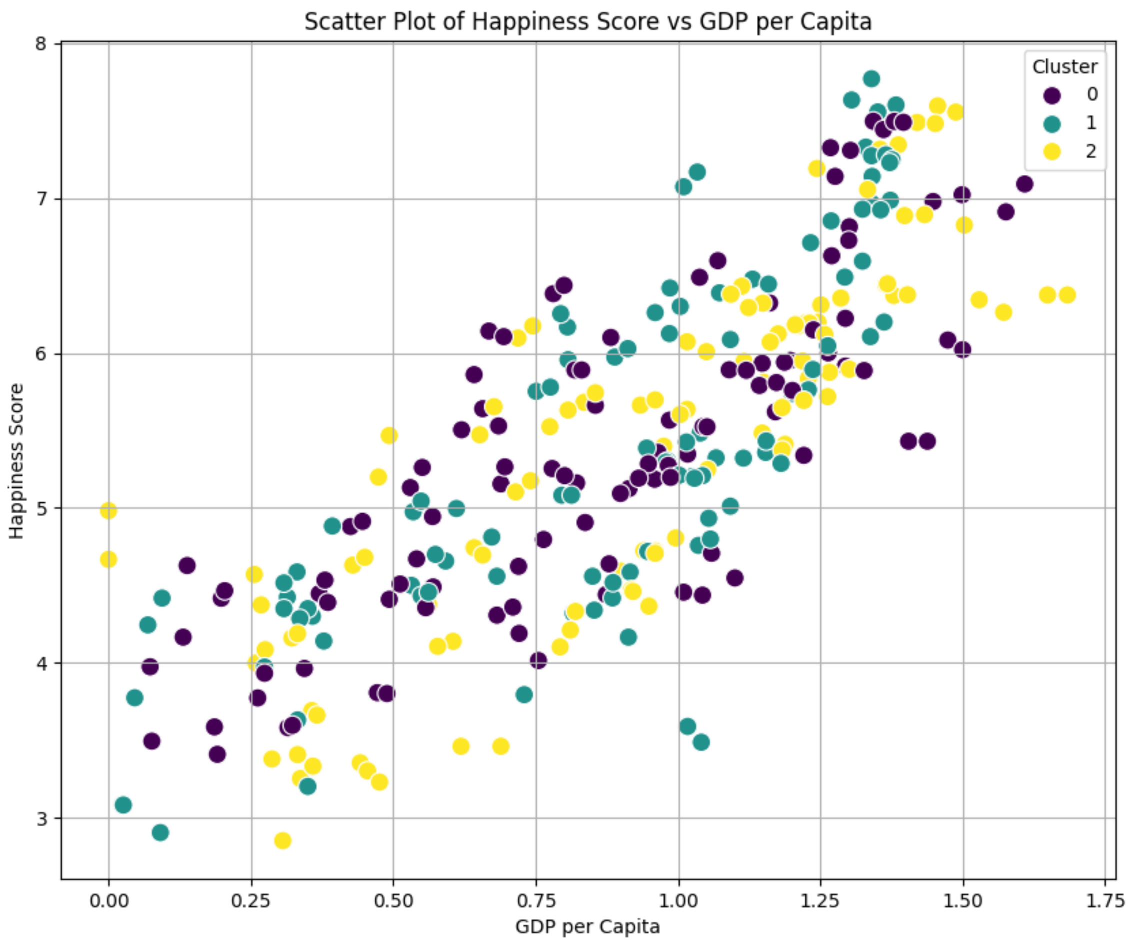
Various visualizations were created to illustrate the correlations and clusters in the data. These visualizations include scatter plots, heatmaps, and cluster plots.



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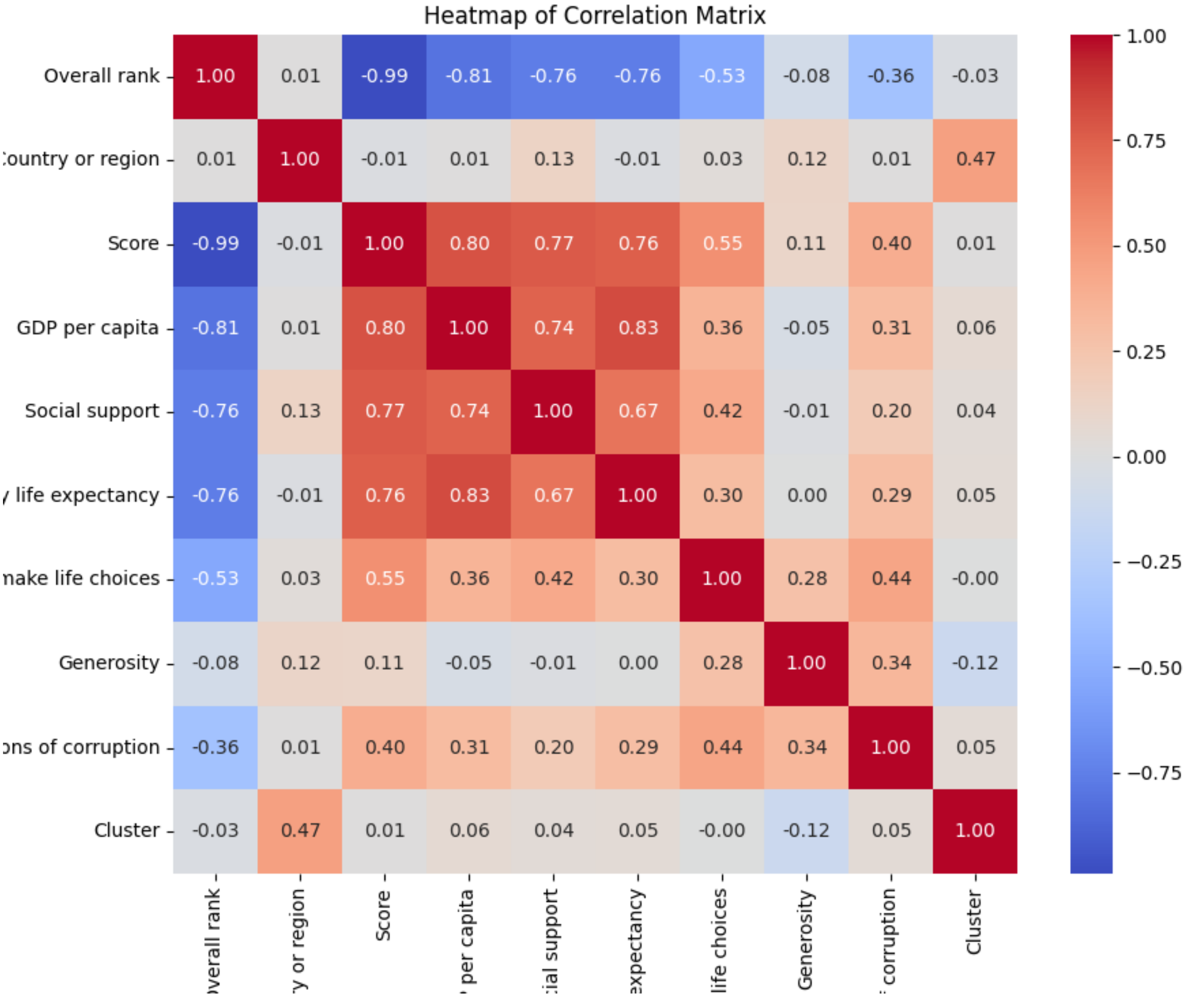

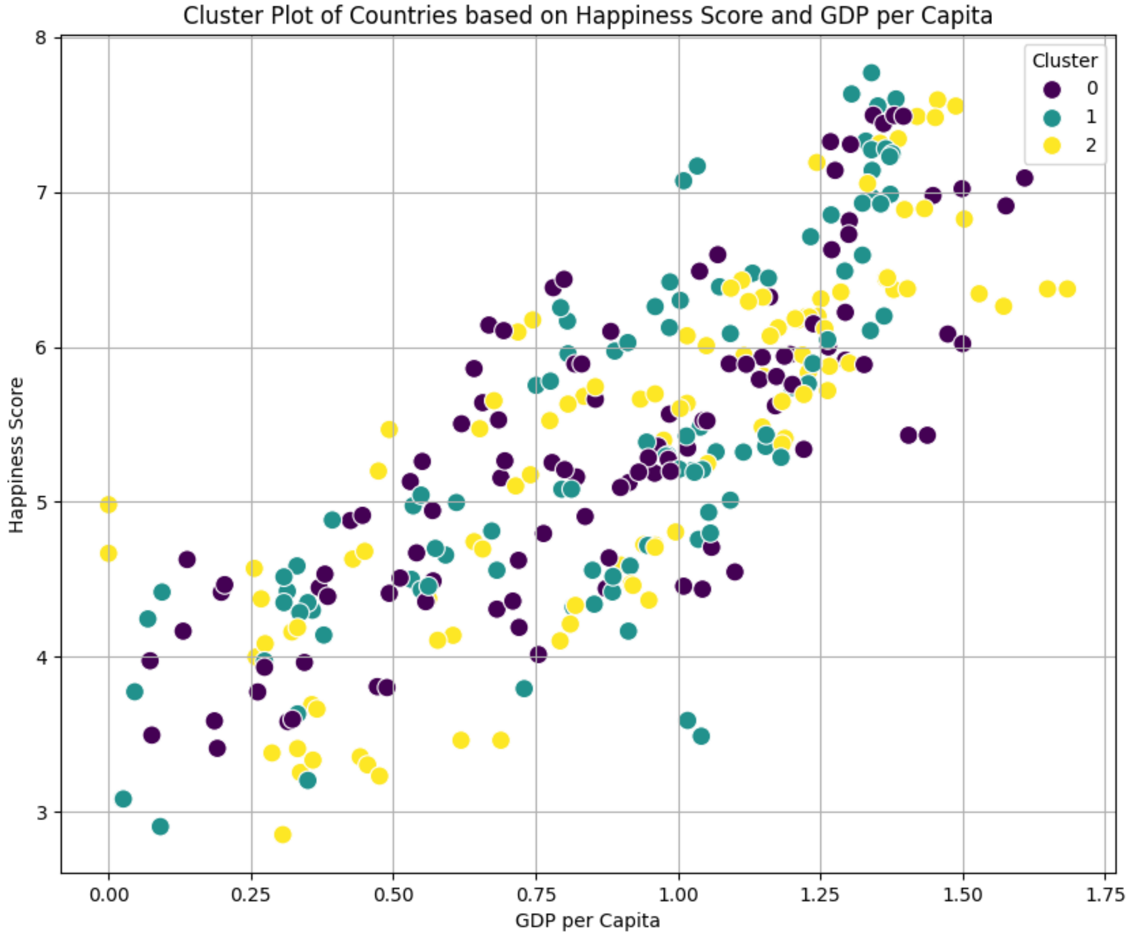

7. Conclusion

The analysis provided deep insights into the factors that contribute to the happiness of nations. The clustering and association rules highlighted the importance of GDP, social support, and health in determining happiness.